CENTRIFUGAL PUMP TECHNICAL DOCUMENTATION

IDENTITY INFORMATION

Company_Name: Meridian Offshore Energy Solutions LLC

Field_Engineer: James R. Patterson

Field_Engineer_Id: MOS-ENG-4782

Document_Version: 2.3.1

Machine_Name: Primary Production Fluid Transfer System (PPFTS-North)

Machine_Id: MOS-PPFTS-N-001

Installation_Date: 2022-06-15

Number_Of_Units_In_Machine: 3

UNIT 1 SPECIFICATIONS

Unit_Name: Centrifugal Pump - Primary Stage

Unit_Id: MOS-CP-PS-0847

Manufacture_Date: 2022-04-22

Manufacturer_Company_Name: Apex Fluid Dynamics Corporation

Manufacturer_Company_Id: AFDC-TX-2891

Inventory_Quantity: 2 units in stock

Supplier_Information: Apex Fluid Dynamics, Houston TX | Contact: procurement@apexfluiddynamics.com | Lead Time: 6-8 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 8,500 feet TVD

- **Operating_Pressure_PSI:** 2,400 PSI nominal / 2,800 PSI maximum

- **Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds (API RP 13B-1 compliant)

- **Known_Error_Codes:** E-CP-001 (cavitation), E-CP-002 (seal failure), E-CP-003 (impeller wear)

- **Replacement_Part_Number:** AFDC-CP-PS-847-REV-C

- **Maintenance_Schedule:** 500-hour intervals; annual full inspection per API RP 54 Section 8.2

UNIT 2 SPECIFICATIONS

Unit_Name: Centrifugal Pump - Secondary Booster Stage

Unit_Id: MOS-CP-BS-0848

Manufacture_Date: 2022-04-22

Manufacturer_Company_Name: Apex Fluid Dynamics Corporation

Manufacturer_Company_Id: AFDC-TX-2891

Inventory_Quantity: 1 unit in stock

Supplier_Information: Apex Fluid Dynamics, Houston TX | Contact: procurement@apexfluiddynamics.com | Lead Time: 6-8 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 8,500 feet TVD

- **Operating_Pressure_PSI:** 1,800 PSI nominal / 2,200 PSI maximum

- **Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds (API RP 13B-1 compliant)

- **Known_Error_Codes:** E-CP-001 (cavitation), E-CP-004 (bearing degradation), E-CP-005 (discharge blockage)

- **Replacement_Part_Number:** AFDC-CP-BS-848-REV-C

- **Maintenance_Schedule:** 500-hour intervals; annual full inspection per API RP 54 Section 8.2

UNIT 3 SPECIFICATIONS

Unit_Name: Centrifugal Pump - Standby/Emergency Unit

Unit_Id: MOS-CP-EM-0849

Manufacture_Date: 2022-05-10

Manufacturer_Company_Name: Apex Fluid Dynamics Corporation

Manufacturer_Company_Id: AFDC-TX-2891

Inventory_Quantity: 0 units in stock (on order)

Supplier_Information: Apex Fluid Dynamics, Houston TX | Contact: procurement@apexfluiddynamics.com | Lead Time: 8-10 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 8,500 feet TVD

- **Operating_Pressure_PSI:** 2,400 PSI nominal / 2,800 PSI maximum

- **Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds (API RP 13B-1 compliant)

- **Known_Error_Codes:** E-CP-001 (cavitation), E-CP-002 (seal failure), E-CP-006 (motor coupling slip)

- **Replacement_Part_Number:** AFDC-CP-EM-849-REV-C

- **Maintenance_Schedule:** 500-hour intervals; annual full inspection per API RP 54 Section 8.2

SECTION 1 - USE CASE

Operational Function and Role:

- **Primary Duty:** Transfer produced fluids (crude oil, water, condensate) from wellhead manifold to separation and storage facilities at flow rates up to 5,000 barrels per day (BPD)

- **System Integration:** Receives inlet pressure from wellhead (200-400 PSI) and boosts to 2,400 PSI for downstream processing and pipeline injection

- **Redundancy Configuration:** Three-unit arrangement provides primary operation (Unit 1), secondary boost capacity (Unit 2), and emergency standby (Unit 3) per API RP 54 Section 4.1

- **Continuous Operation:** Designed for 24/7 operation with scheduled maintenance windows; critical to production continuity and revenue protection

- **Fluid Handling:** Manages multiphase fluids containing dissolved gases, water content, and particulate matter per API RP 13B-1 standards

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SECTION 2 - FAILURE CAUSES

Known Failure Modes (Ranked by Frequency):

- **E-CP-001 Cavitation (35% of failures):** Occurs when inlet pressure drops below vapor pressure; caused by restricted suction lines, low reservoir levels, or excessive flow rates; leads to impeller pitting and noise

- **E-CP-002 Mechanical Seal Failure (28% of failures):** Premature wear from abrasive particulates in drilling fluid; exacerbated by temperature cycling and pressure spikes; results in fluid leakage and environmental risk

- **E-CP-003 Impeller Erosion/Wear (18% of failures):** Progressive material loss from sand production and high-velocity fluid impact; reduces pump efficiency and discharge pressure over 1,500-2,000 operating hours

- **E-CP-004 Bearing Degradation (12% of failures):** Lubrication breakdown under sustained high-pressure operation; accelerated by contaminated cooling water and thermal cycling

- **E-CP-005 Discharge Line Blockage (7% of failures):** Paraffin deposition, scale formation, or debris accumulation in downstream piping; causes pressure relief activation and reduced throughput

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SECTION 3 - SOLUTIONS

Corrective Actions (API RP 54 & OSHA 1910 Compliant):

- **Cavitation Prevention (E-CP-001):** Maintain inlet pressure ≥50 PSI above vapor pressure; inspect suction strainers weekly; verify reservoir level minimum 6 feet; reduce flow rate if inlet pressure drops below threshold per OSHA 1910.119(e) process safety management

- **Seal Replacement Protocol (E-CP-002):** Replace mechanical seals every 1,000 operating hours or upon visual leakage; use API Plan 53A flush system with compatible barrier fluid; perform replacement during scheduled maintenance windows with proper lockout/tagout (LOTO) per OSHA 1910.147

- **Impeller Inspection & Replacement (E-CP-003):** Conduct ultrasonic thickness measurements at 500-hour intervals; replace impeller when wear exceeds 3mm radial clearance; use OEM replacement parts (AFDC-CP-PS-847-REV-C) to maintain performance curves

- **Bearing Maintenance (E-CP-004):** Change bearing cooling water every 250 hours; maintain water temperature 40-60°C; monitor bearing temperature with infrared thermography; replace bearings if temperature exceeds 80°C per API RP 54 Section 8.2.3

- **Discharge Line Flushing (E-CP-005):** Perform hot oil circulation flush monthly; install 100-micron discharge strainer; monitor differential pressure across strainer; clean or replace strainer element when ΔP exceeds 15 PSI per API RP 13B-1

SECTION 4 - INVENTORY DATA

Stock Management and Procurement:

- **Current Stock Levels:** Unit 1 (Primary Stage): 2 units available | Unit 2 (Booster Stage): 1 unit available | Unit 3 (Emergency): 0 units (on order, ETA 8-10 weeks)

- **Reorder Thresholds:** Maintain minimum 1 unit per pump type in stock; trigger reorder when inventory drops below threshold; critical spare parts (seals, impellers, bearings) stocked at 5-unit quantities

- **Supplier & Lead Times:** Primary supplier Apex Fluid Dynamics (AFDC-TX-2891); standard lead time 6-8 weeks for complete pump assemblies; expedited orders available at 15% premium for 2-3 week delivery

- **Annual Procurement Budget:** Estimated $185,000 USD for replacement units and consumable maintenance parts; budget includes seal kits ($2,400/unit), impeller assemblies ($8,900/unit), and bearing sets ($3,200/unit)

Document Prepared By: James R. Patterson, Field Engineer MOS-ENG-4782

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Compliance Status: ✓ OSHA 1910 | ✓ API RP 54 | ✓ API RP 13B-1

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